Department of Scientific Computing, Modeling and Simulation, SPPU

AI/ML Project Synopsis

AI Research Paper Assistant using Agentic RAG

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Introduction

The AI Research Paper Assistant uses Agentic Retrieval-Augmented Generation (RAG) to help researchers efficiently analyse research papers. Users can upload multiple PDFs, ask questions, compare papers, generate literature reviews, summaries, flashcards, quizzes, citations, and identify research gaps using Large Language Models and semantic search.

Problem Statement

Traditional literature review requires significant manual effort to analyse multiple papers, identify important findings, compare research, and prepare surveys. Existing tools lack integrated AI-based document understanding and comparison capabilities.

Objectives

- Develop an Agentic RAG assistant for research papers.
- Enable Multi-PDF conversational question answering.
- Generate literature reviews, summaries, citations, flashcards and quizzes.
- Compare papers and identify research gaps.
- Build a knowledge graph across papers.
- Export reports as PDF/Word.

Methodology

Research papers are parsed into text chunks, converted into embeddings, and stored in a vector database. Relevant information is retrieved using semantic search and passed to an LLM. AI agents perform literature review generation, comparison, citation generation, research gap identification, and knowledge graph creation.

Project Phases

Phase 1: Requirement Analysis	**Phase 2:** System Design
Phase 3: Frontend Development	**Phase 4:** Backend Development
Phase 5: RAG Integration	**Phase 6:** AI Agent Development
Phase 7: Testing	**Phase 8:** Deployment

Technologies Used

Python, React.js, FastAPI, LangChain, LangGraph, OpenAI/Llama Models, Hugging Face, FAISS/ChromaDB, PyMuPDF, PostgreSQL/MongoDB, JWT Authentication, ReportLab, python-docx, Whisper API.

Expected Outcomes

The system will provide intelligent document-based question answering, literature review generation, paper comparison, research gap identification, citation generation, and personalised research assistance, improving academic productivity and research efficiency.